STAT 405 Final Project Proposal

1. Team Information

Team member:

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2. Research Question

This project applies Bayesian inference to baseball batting performance. We will use the **500hits** dataset, which contains career batting statistics for 463 baseball players. In particular, we will focus on the variables **AB** (total at-bats) and **H** (total hits).

Our main scientific question is: **Which player has the highest underlying hitting probability after accounting for uncertainty, and how do player rankings change under different Bayesian models and posterior computation methods?**

3. Candidate Datasets

Two candidate datasets have been identified for this project.

Main dataset: Baseball Player Metrics

Our main dataset is the **Baseball Player Metrics / 500hits** dataset. It contains career-level statistics for 463 baseball players and includes variables such as years played, games, at-bats, runs, hits, doubles, triples, home runs, and RBIs.

For this project, we will primarily use:

- Player: player name
- YRS: number of years active in professional baseball
- G: total games played
- AB: total at-bats
- H: total hits
- BA: observed batting average

This dataset is appropriate for our project because for each player, the number of hits can naturally be modelled as a Binomial outcome with the number of trials equal to at-bats. This gives a clean and interpretable Bayesian modeling framework for estimating player-specific hitting probabilities.

Here is the structure of the dataset:

```
##             PLAYER YRS    G    AB    R    H X2B X3B  HR  RBI   BB   SO  SB  CS
## 1          Ty Cobb  24 3035 11434 2246 4189 724 295 117  726 1249  357 892 178
## 2      Stan Musial  22 3026 10972 1949 3630 725 177 475 1951 1599  696  78  31
## 3     Tris Speaker  22 2789 10195 1882 3514 792 222 117  724 1381  220 432 129
## 4      Derek Jeter  20 2747 11195 1923 3465 544  66 260 1311 1082 1840 358  97
## 5     Honus Wagner  21 2792 10430 1736 3430 640 252 101    0  963  327 722  15
## 6 Carl Yastrzemski  23 3308 11988 1816 3419 646  59 452 1844 1845 1393 168 116
##      BA HOF
## 1 0.366   1
## 2 0.331   1
## 3 0.345   1
## 4 0.310   1
## 5 0.329   1
## 6 0.285   1
```

Source:

https://www.kaggle.com/datasets/krupadharamshi/500hits/data

Dataset 2 (Backup Dataset)

Dataset name:

Source:

https://archive.ics.uci.edu/dataset/45/heart+disease

This dataset contains

If necessary, this dataset can be used to conduct a similar Bayesian analysis of

4. **Proposed Models**

We propose to analyze the data using two Bayesian models.

Model 1:

The first model is a Bayesian

Model 2:

5. **Posterior Computation Methods**

The posterior distribution will be approximated using two different methods.

Method 1:

Method 2:

6. **Model Comparison and Evaluation**

We will compare the models and inference methods using several criteria.

These include:

posterior means and credible intervals

convergence diagnostics

predictive performance

posterior predictive checks

This comparison will help evaluate how model complexity and inference methods affect the estimated relationships.

7. **Implementation Plan**

The project will proceed in the following steps:

1. Data cleaning and exploratory data analysis

2. Implementation of Bayesian
3. Posterior inference using
4. Posterior inference using
5. Model comparison and diagnostic analysis

6. Preparation of the final report

8. **Contribution Plan**

We will divide the work equally. One team member will focus on data cleaning, exploratory analysis, and implementation of the baseline no-pooling model. The other team member will focus on the hierarchical model, posterior computation. We will do model comparison together. Both team members will contribute to writing, interpretation, and final editing.

9. **Git Repository**

The code for this project will be maintained in a public Git repository.

Repository link:

The repository will include data preprocessing scripts, model implementation code, and scripts for generating figures and tables used in the report.